

Causal Trust Region Optimization: Protecting Latent Geometry under Policy Gradients

Early draft

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Abstract

Policy-gradient methods with KL trust regions (TRPO, PPO) constrain how much the *action distribution* may change, yet they do not directly stabilize the latent features the policy uses. Prior work shows those features can still collapse in rank under PPO. We argue that what ultimately matters for generalization is a well-formed *causal latent* representation Z of the state, and that local guarantees relating policy updates to representation error require the value function V to retain μ -like strong convexity structure *in* Z . We introduce an operational method—**Causal Trust Region Optimization** (CTRO)—that trains a shared encoder so both policy and critic act on Z , and adds two regularizers: a bisimulation-style metric loss (MICo) that anchors Z in reward–transition geometry, and a Polyak–Łojasiewicz (PL) coupling that keeps value geometry over Z from degenerate collapse. Empirically, on Minigrid ablations and on Procgen and DeepMind Control performance suites, full CTRO improves average return over plain PPO and over a matched shared-latent baseline without the extra losses, and raises a online μ_{PL} diagnostic. Feature rank alone does not separate good from bad cells in our ablations. A direct latent trust region D_Z remains concrete future work.

1 Introduction and motivation

Proximal Policy Optimization and Trust Region Policy Optimization are standard workhorses of deep reinforcement learning. They stabilize learning by bounding KL divergence between successive policies, which usually prevents catastrophic policy updates. Yet the object that matters for generalization is often not only $\pi(a \mid s)$, but the internal state summary $Z(s)$ that the network uses when choosing actions and estimating values.

Two lines of theory and evidence motivate this work.

Causal representations. Richens and Everitt [1] argue that, under local distributional shifts, low-regret decision making essentially requires an approximate *causal* model of the environment. In deep RL, such a model is not given explicitly; it must emerge as a learned latent $Z(s)$. A natural research goal follows: stabilize and identify a latent that can play the role of a causal state for both policy and value.

Why KL is not enough. Moalla et al. [2] (and related work on feature collapse, e.g. Lyle et al. [3]) show that hidden features under PPO can still lose effective rank even when policy updates are KL-constrained. Rank collapse is a capacity pathology; more generally, the KL only bounds change of *action distributions*. It is a loose proxy for stability of Z when the mapping $s \mapsto Z(s)$ and the heads $\pi(\cdot \mid Z)$ and $V(Z)$ co-train.

Value geometry in latent space. If the critic V is locally μ -strongly convex in Z about a well-chosen optimum Z^* , representation error and the gradient $\nabla_Z V$ are tightly related. Under that structure, one can form a chain of bounds linking policy KL, value-gradient change, latent displacement, and regret (informal summary in §2). When that μ -structure fails, the chain breaks and KL clipping no longer “protects” the representation.

What this paper does. We do *not* yet replace PPO’s KL trust region with a proven latent constraint. Instead we propose an operational CTRO recipe that (i) makes Z the shared input of π and V , and (ii) regularizes Z so that value geometry stays healthy and the latent stays metrically tied to reward and transition structure. A latent distance trust region D_Z is outlined as future work (§7).

2 Theoretical backbone (informal)

We sketch the reasoning that shapes the method. A full formalization is out of scope for this draft.

Step 1 — Stabilizing Z , not only π . If almost all hard transfer tasks require an approximate causal model [1], the right object to control is $Z^*(s)$, not only the action law $\pi(\cdot | s)$.

Step 2 — Representation error and $\nabla_Z V$. If V is μ -strongly convex in Z near Z^* with $\nabla_Z V(Z^*) = 0$, then roughly

$$\|Z^*(s) - Z(s)\| \leq \frac{1}{\mu} \|\nabla_Z V(Z(s))\|. \quad (1)$$

Intuitively: V is a bowl in latent space; the slope measures distance to the bottom when the bowl is tightly curved (large μ).

Step 3 — Policy change bounds gradient change. Via policy-gradient structure and inequalities relating total variation to KL, successive policies at small KL differ in value gradients by order $\sqrt{\epsilon_{\text{KL}}}$ (plus vanishing parameter-step terms under a trust region). Details omitted; see theory notes.

Step 4 — Chain. Combining these yields an informal *causal-policy bounding chain*: regret controlled by representation error, controlled by value-gradient mismatch, controlled by policy KL—*when* μ -structure holds. **Failure mode:** if V is flat or concave along directions of Z , KL-bounded updates no longer constrain Z in a way that the bowl argument cares about.

What theory demands of an algorithm. A complete method would (a) keep μ -structure of V on Z alive, (b) identify Z so it is not an arbitrary gauge for fitting V , and (c) optionally constrain drift $\|Z_{\theta'} - Z_{\theta}\|$ directly (D_Z). Part (c) is future work; parts (a)–(b) are implemented as the two regularizers below.

3 Method: CTRO

3.1 Shared latent

Let e_ϕ be an encoder, $Z = e_\phi(s)$. Both the policy and critic use Z :

$$\pi_\theta(a \mid Z), \quad V_\psi(Z).$$

This is the **architecture triad**: encoder, value head, policy head. The encoder is trained by backprop through the usual PPO losses and through CTRO terms, so representation learning is jointly driven by control and value, not only reconstruction.

Domain stacks.

- **Pixels (Procggen / Minigrid)**: convolutional (or Impala-style) encoder maps images to Z ; optional VAE path for some Minigrid runs only.
- **State control (DMControl)**: MLP encoder maps compact physics state to Z (no VAE). Compact Markov states do not justify a generative bottleneck.

3.2 Objective

CTRO extends PPO with two additive losses:

$$L_{\text{CTRO}} = L_{\text{PPO}} + \alpha L_{\text{MICo}} + \beta L_{\text{PL}}. \quad (2)$$

Policy updates remain standard PPO (clip / KL-proximal). We do *not* use a representation-space constraint in the current optimizer.

MICo (causal / metric pin). L_{MICo} is a bisimulation-style (MICo) metric loss on pairs of states: latent distances should track absolute reward differences plus discounted next-latent distances. Role: reduce *gauge freedom* of Z with respect to dynamical/reward structure. Without it, one can reparameterize Z for the value head’s convenience while scrambling causal neighborhoods.

PL coupling (value-geometry pin). L_{PL} is a hinge on the online PL ratio of the value head over Z ,

$$\mu_{\text{PL}} = \frac{\|\nabla_Z V(Z)\|^2}{2 \cdot \text{gap}}, \quad (3)$$

with gap a one-step Bellman residual proxy. We penalize μ_{PL} falling below a floor μ_0 . Role: keep value geometry non-degenerate on the shared latent—the operational substitute for maintaining μ -bowl structure.

Division of labor.

Term	Role
Shared $Z + \pi(\cdot \mid Z)$, $V(Z)$	Same coordinates for actor and critic.
MICo (α)	Anchor Z in reward–transition metric (causal gauge).
PL (β)	Keep value geometry healthy (μ_{PL}).
PPO KL/clip	Standard policy trust region (proxy only for Z).

Neither pin replaces the other: structure without value link is insufficient; value link without metric anchoring leaves Z under-identified.

3.3 Diagnostics

μ_{PL} . Logged during training (e.g. 5th percentile across batch samples). Tracks the value-geometry pin.

Participation ratio (PR). Effective feature rank proxy of the latent covariance (Moalla/Lyle-style capacity measure). Used as a secondary diagnostic, not the primary success metric.

4 Experimental setup

4.1 Minigrid 2×2 loss ablation

Fast setting with fixed VAE + latent policy stack. Four cells over $\{\text{MICo off/on}\} \times \{\text{PL off/on}\}$, three seeds. Isolates *losses*, not architecture. Primary tables: final return, μ_{PL} , PR.

4.2 Performance suites with negative controls

Three conditions on every task (seeds 42–44):

Condition	Role
Baseline	Plain PPO on domain-standard architecture. External performance reference.
Latent, no link	Same shared Z as CTRO, π and V on Z , but $\alpha = \beta = 0$. Architecture without CTRO losses.
CTRO	Shared Z + MICo + PL.

Progen easy (8 games: caveflyer, chaser, coinrun, fruitbot, leaper, maze, miner, starpilot). CTRO: CNN encoder $\rightarrow Z$ (`exp_ctro_cnn`).

DMControl state (4 tasks: cartpole-swingup, cheetah-run, hopper-hop, walker-walk). CTRO: MLP encoder $\rightarrow Z$ (`exp_ctro_mlp_v2`).

Metrics. Primary: held-out evaluation returns after training (mean $\pm$ seed std). Secondary: online μ_{PL} and PR from training logs.

4.3 Hypotheses

H1

Full CTRO improves average return versus baseline PPO.

H2

Shared latent without MICo/PL is not enough (latent-no-link underperforms full CTRO).

H3

Full CTRO raises μ_{PL} relative to controls; PR need not cleanly separate methods.

5 Results

5.1 Minigrid ablation (summary)

From the controlled 2×2 (mean over three seeds): cells with PL on raise both μ_{PL} and return; MICO alone does not (structure without value link). $PR \approx 1$ in every cell—rank collapse does not discriminate success from failure here. This supports treating value geometry, not feature rank, as the primary empirical axis for CTRO.

5.2 Performance suites: average return

Table 1: Mean held-out `full_return` across tasks (aggregated 5–6 Aug 2026). Bold is best.

Suite	Baseline	Latent, no link	CTRO
Procgen easy (8 games)	4.85	5.23	7.56
DeepMind Control (4 tasks)	14.6	170	285

On average, $CTRO \succ \text{latent-nolink} \succ \text{baseline}$ (H1, H2). Gains are large on DeepMind Control. On Procgen the mean still favors CTRO, with uneven per-game outcomes.

5.3 DeepMind Control

Table 2: DMControl `full_return` (mean $\pm$ std over seeds). $CTRO = \text{exp_ctro_mlp_v2}$.

Task	Baseline	Latent, no link	CTRO
cartpole-swingup	25 ± 35	387 ± 43	597 ± 257
cheetah-run	4.3 ± 5.2	148 ± 63	396 ± 19
hopper-hop	0.11 ± 0.09	0.10 ± 0.11	0.35 ± 0.03
walker-walk	29 ± 1	146 ± 70	147 ± 37
Mean	14.6	170	285

Cartpole and cheetah carry most of the signal. Walker is similar for CTRO and latent-no-link (both far above baseline). Hopper remains near zero for all methods under this budget—an unsolved task rather than a separator.

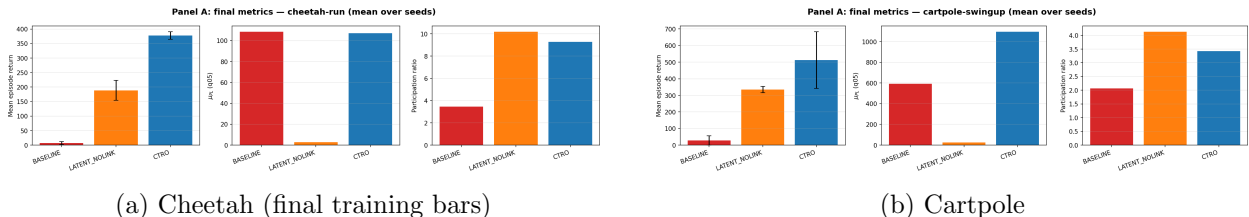


Figure 1: Illustrative DMControl final-return comparisons.

5.4 Procgén easy

Table 3: Procgén `full_return` (mean $\pm$ std). CTRO = `exp_ctro_cnn`.

Task	Baseline	Latent, no link	CTRO
caveflyer	1.38 ± 0.34	1.27 ± 1.30	3.54 ± 1.43
chaser	1.06 ± 0.10	1.02 ± 0.70	1.31 ± 0.17
coinrun	7.37 ± 0.29	3.53 ± 2.68	4.93 ± 1.16
fruitbot	17.95 ± 2.24	16.79 ± 12.64	23.43 ± 0.97
leaper	3.93 ± 0.54	3.33 ± 0.59	3.70 ± 0.62
maze	3.53 ± 2.49	1.07 ± 0.12	3.00 ± 0.62
miner	0.52 ± 0.38	0.10 ± 0.08	0.20 ± 0.02
starpilot	3.07 ± 0.53	14.72 ± 10.41	20.33 ± 3.61
Mean	4.85	5.23	7.56

CTRO leads on fruitbot and starpilot and on the suite mean. Baseline remains strongest on some games (coinrun, miner). Latent-no-link is often high-variance—shared Z alone is not a reliable recipe (H2).

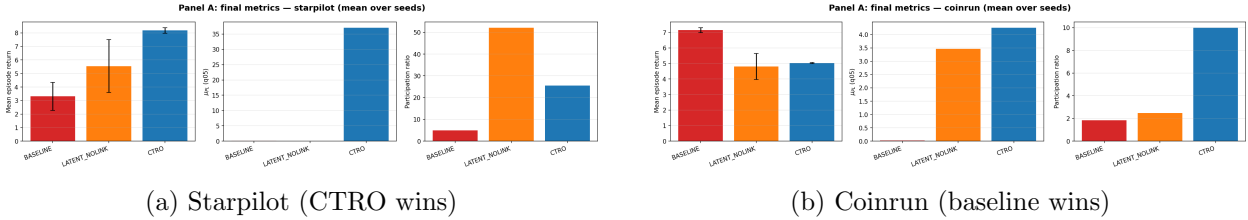


Figure 2: Procgén games illustrating mixed per-task outcomes under a positive suite mean for CTRO.

5.5 Geometry (H3)

Across both suites, full CTRO reliably elevates online μ_{PL} relative to baseline. Latent-no-link often shows weak μ_{PL} even when return is intermediate. Participation ratio often rises under CTRO but is not a clean separator: some latent-no-link runs show high PR with weak μ_{PL} and unstable return. This is consistent with the Minigrid message: **value geometry tracks the intervention better than rank**.

5.6 Reading against the three hypotheses

H1

Supported on suite averages (Procgén and DMControl).

H2

Strongly supported on DMControl; on Procgén, full CTRO still has the best mean, with game-level exceptions and architecture confounds for baseline vs matched stacks.

H3

Supported directionally for μ_{PL} ; not for PR as a unique separator.

6 Discussion

What we claim. CTRO is *value-linked, metrically anchored latents under PPO*. It operationalizes the idea that both π and V should live on a causal-style Z , and that keeping value geometry (μ_{PL}) and bisimulation metric structure (MICo) protects performance beyond “just share a latent.”

What we do not claim.

- We do *not* claim to disprove Moalla et al. We often *see* rank collapse; it simply does not explain our method gaps.
- We do *not* claim a certified μ -strong convex V or a closed contraction lemma for joint on-policy training.
- We do *not* yet run a representation-space trust region D_Z .
- Per-task stories remain noisy with three seeds; hopper is unsolved for everyone in the current suite.

Relation to earlier designs. Earlier program material (ICNN critics, Hessian $\lambda_{\min}$ monitoring, RSTR sketches) targets the same μ -bowl premise. The live method substitutes **PL as a practical μ proxy** and **MICo as an identification / gauge pin**, under standard PPO, rather than completing a Fisher-constrained D_Z step.

7 Future work: a latent trust region D_Z

A natural next step—suitable as a follow-on paper or a strengthened extension—is to *constrain latent drift directly* during policy or encoder updates:

$$D_Z(\theta', \theta) = \mathbb{E}_{s \sim d_\pi} \|Z_{\theta'}(s) - Z_\theta(s)\|^2 \leq \delta_Z, \quad (4)$$

or a local quadratic form via a representation Fisher F_Z (TRPO-style step in Z -coordinates).

Why it could help. D_Z enforces the missing piece of the bounding chain: small policy steps imply small $\|Z' - Z\|$ by design, not only via KL. It would pair with CTRO’s soft geometry losses rather than replace them: MICo and PL identify a good Z ; D_Z keeps optimization from jumping out of the local μ -neighborhood.

What it needs.

- A practical estimator of F_Z or a cheaper soft penalty on $\|Z' - Z\|^2$.
- Care about which parameters enter θ (encoder, policy-on- Z , critic phasing).
- Soft target latents to avoid a moving-target pathology.
- Ablations against PPO+CTRO alone to show additive value.

We leave a complete D_Z algorithm as **concrete future work**, not a claimed contribution of this draft.

8 Conclusion

We linked three ideas: causal latents as necessities for robust policies; μ -structure of V on Z as the condition that makes trust-region reasoning about representations meaningful; and a practical CTRO recipe that shares Z across π and V while pinching causal metric structure (MICo) and value geometry (PL). On Minigrid, Procgen, and DeepMind Control, full CTRO improves average returns and μ_{PL} relative to controls, while feature rank alone fails as a method separator. A latent D_Z trust region is a clear next algorithmic step.

Draft status. Numbers are from the August 2026 experiment log. Citations below are placeholders for full bibliography cleanup.

References

- [1] J. Richens and T. Everitt. Robust agents learn causal world models. *ICLR*, 2024.
- [2] S. Moalla, A. Miele, R. Pascanu, and C. Gulcehre. No representation, no trust: Connecting representation, collapse, and trust regions. arXiv preprint, 2024.
- [3] C. Lyle, M. Rowland, and W. Dabney. Understanding and preventing capacity loss in reinforcement learning. *ICLR*, 2022.